

1 Highlights

2 **Water transparency retrieval in a high-altitude large lake from** 3 **Sentinel-2 and Landsat 8/9: cross-sensor harmonization limits, val-** 4 **idation design, and calibrated uncertainty in Lake Titicaca**

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- 7 • Land-derived Landsat/Sentinel-2 harmonization is invalid over dark
8 oligotrophic water
- 9 • Sensors stay 99.8% separable after harmonization; intercepts exceed
10 signal 70-fold
- 11 • Campaign, not station, is the blocking unit in campaign-based moni-
12 toring programmes
- 13 • Honest Secchi retrieval: $R^2=0.57$, RMSE 1.86 m over 812 real match-
14 ups
- 15 • Conformal intervals are calibrated marginally but under-cover in the
16 clearest water

17 Water transparency retrieval in a high-altitude large
18 lake from Sentinel-2 and Landsat 8/9: cross-sensor
19 harmonization limits, validation design, and calibrated
20 uncertainty in Lake Titicaca

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23 **Abstract**

24 Lake Titicaca (8372 km², 3810 m a.s.l.) is the largest lake in South Amer-
25 ica and a transboundary freshwater reserve for millions of people, yet its
26 water quality is documented only by infrequent field campaigns. We assem-
27 bled 1002 satellite–field match-ups (340 Sentinel-2, 662 Landsat 8/9) against
28 twelve years of IMARPE limnological monitoring and used them to ask what
29 a defensible multi-mission Secchi disk depth (Z_{SD}) product looks like in clear
30 oligotrophic high-altitude water. Two widely used practices fail here. First,
31 the cross-sensor harmonization coefficients of Roy et al. (2016), fitted over ter-
32 restrial targets and routinely reused to build harmonized Landsat/Sentinel-2
33 archives, are invalid over dark water: their additive intercepts (0.0116–0.0448
34 in reflectance) exceed the native over-water Landsat signal by factors of 1.4
35 to 69.8, and a classifier still identifies the source sensor with 99.8% accuracy
36 after the transform (chance 66.3%). Refitting locally on 270 same-day paired
37 observations restores agreement in green ($r = 0.71$) but not in blue ($r = 0.26$)
38 — the band the retrieval most depends on. Second, blocking cross-validation

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39 by monitoring station, the usual remedy for spatial autocorrelation, retains
 40 nothing in a campaign-based programme: under campaign-blocked folds a
 41 station-mean null still reaches $R^2 = 0.208$ and a zone-mean null $R^2 = 0.287$,
 42 while the campaign-mean null collapses to -0.012 . Under campaign-blocked
 43 validation a Random Forest retrieves Z_{SD} with $R^2 = 0.574$ and RMSE 1.86 m
 44 ($n = 812$), against 0.097 for a correctly bounded classical blue/green band
 45 ratio. Split-conformal intervals are well calibrated marginally (89.4% empir-
 46 ical coverage at 90% nominal) but not conditionally, under-covering in the
 47 clearest water (71.3%). Skill collapses under leave-one-zone-out extrapola-
 48 tion, and chlorophyll- a , suspended solids and thermal temperature are not
 49 retrievable. We propose two inexpensive diagnostics — a sensor classifier
 50 and null models under each candidate blocking unit — that separate genuine
 51 optical retrieval from sensor labelling and sampling design.
 52 *Keywords:* Lake Titicaca, Secchi disk depth, Sentinel-2, Landsat,
 53 cross-sensor harmonization, conformal prediction, large lakes

54 1. Introduction

55 Large lakes are sentinels of environmental change, but continuous moni-
 56 toring of remote, high-altitude systems is constrained by sparse field infras-
 57 tructure (Williamson et al., 2009). Lake Titicaca, shared by Peru and Bolivia
 58 at 3810 m a.s.l., is the largest lake in South America and, by volume, the
 59 largest tropical lake in the world. It is a Ramsar site and the freshwater
 60 reserve for millions of Andean people, yet its water quality is documented
 61 only by infrequent field campaigns (OEFA, 2016; Siguayro Mamani et al.,
 62 2022), even as eutrophication advances in its near-shore bays (Heredia et al.,

63 2022; Lanza et al., 2024).

64 Satellite remote sensing offers synoptic, repeatable observation of large
65 lakes (Gholizadeh et al., 2016; Topp et al., 2020). Water transparency, quan-
66 tified by the Secchi disk depth (Z_{SD}) since Tyler (1968), is an integrative
67 indicator of water clarity strongly linked to the diffuse attenuation of light
68 (Lee et al., 2015), and is among the most reliably retrievable optical water-
69 quality variables; it has been mapped operationally from Landsat for two
70 decades (Kloiber et al., 2002; Olmanson et al., 2008; Zhang et al., 2022).
71 Sentinel-2 MSI (10–60 m) and Landsat 8/9 OLI (30 m) provide complemen-
72 tary revisit, and their joint use as a virtual constellation is now standard
73 practice (Claverie et al., 2018; Pahlevan et al., 2019). That joint use rests on
74 cross-sensor harmonization, for which the per-band linear coefficients of Roy
75 et al. (2016) are the most widely reused transform. Machine learning applied
76 to such harmonized archives dominates water-quality retrieval across many
77 lakes (Pahlevan et al., 2020; Cao et al., 2020, 2022; Arias-Rodriguez et al.,
78 2023; Pahlevan et al., 2022; Hafeez et al., 2019; Tian et al., 2023).

79 This literature is overwhelmingly built on turbid, productive “Case-2”
80 waters, where chlorophyll and mineral turbidity carry a strong optical sig-
81 nal (Gitelson et al., 2009; Mishra and Mishra, 2012; Nechad et al., 2010;
82 Dogliotti et al., 2015). Clear oligotrophic high-altitude lakes are compara-
83 tively neglected, and two assumptions imported from that literature have not
84 been tested in them.

85 The first is that harmonization coefficients derived over land transfer to
86 water. The Roy et al. (2016) transforms were fitted on terrestrial targets —
87 vegetation, soil and built surfaces — whose reflectance is one to two orders of

88 magnitude brighter than clear water. Their additive intercepts are therefore
 89 of the same order as, or larger than, the entire over-water signal. That
 90 the transform degrades over water is not a new suspicion: a positive near-
 91 infrared bias over dark targets has been noted (Pahlevan et al., 2019), and the
 92 HLS product documentation itself cautions that its atmospheric correction
 93 is not optimized for water and that water reflectance is unreliable (Masek
 94 et al., 2021; Claverie et al., 2018). What is missing is quantification. To
 95 our knowledge no study has measured how large the mismatch is for a clear
 96 oligotrophic lake, shown what it does to a machine-learning retrieval trained
 97 on the pooled archive, or reported a routine check that would catch it.

98 The second is that cross-validation blocked by monitoring station is suf-
 99 ficient to demonstrate generalization. Accuracy in this field is routinely re-
 100 ported under random train/test splits that ignore spatial structure, a prac-
 101 tice known to inflate apparent skill when samples are spatially autocorrelated
 102 (Roberts et al., 2017; Ploton et al., 2020; Meyer et al., 2019). Blocking by
 103 station is the usual remedy. It is the right remedy only if station identity is
 104 where the dependence lives — an assumption that is rarely tested and that,
 105 we show, fails in a campaign-based monitoring programme.

106 For Lake Titicaca specifically, prior remote-sensing work has used MODIS/Landsat
 107 temperature and chlorophyll (Ruiz-Verdú et al., 2016), basin-scale land-cover
 108 drivers (Baltodano et al., 2022), automatic Sentinel-2 water products (Ay-
 109 ala Izurieta et al., 2023), and global remote-sensing water-quality products
 110 (Maligaya et al., 2024). None provides a harmonized multi-mission Z_{SD}
 111 retrieval validated against in-situ measurements at their true sampling coor-
 112 dinates, and none assesses which variables are actually retrievable in its clear

113 water.

114 We address these gaps with 1002 real satellite–field match-ups. Our con-
115 tributions are threefold. First, we quantify the failure of land-derived cross-
116 sensor harmonization over clear oligotrophic water, derive water-specific re-
117 placement coefficients from same-day paired observations, and adapt an es-
118 tablished statistical test — the classifier two-sample test (Friedman, 2003;
119 Lopez-Paz and Oquab, 2017), known in applied machine learning as adver-
120 sarial validation — into a one-line harmonization check that any study can
121 run. Second, we show that the blocking unit for validation must be diagnosed
122 rather than assumed, and give the null-model procedure for doing so. Third,
123 we report retrieval accuracy, calibrated per-prediction uncertainty and an
124 explicit retrievability envelope under the resulting design. Our aim is not to
125 claim a high-accuracy product but to establish what a defensible one looks
126 like for this class of lake.

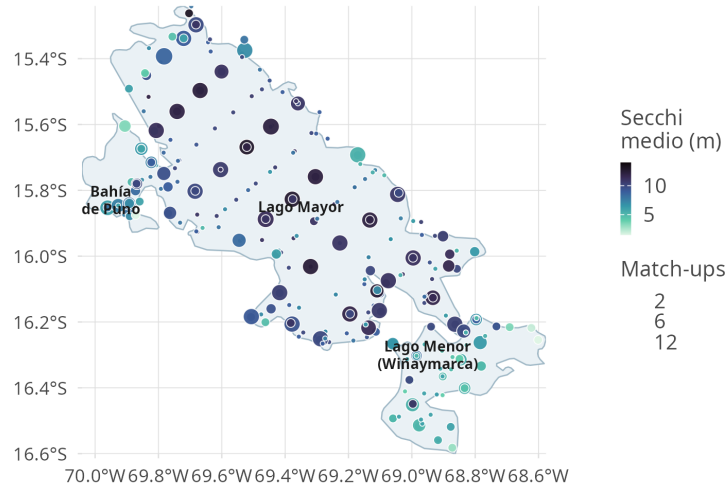
127 **2. Materials and methods**

128 *2.1. Study area*

129 Lake Titicaca comprises three limnologically distinct zones (Fig. 1): the
130 large, deep, oligotrophic Lago Mayor; the shallower Lago Menor (Wiñay-
131 marca) to the south-east; and Bahía de Puno, which receives municipal and
132 agricultural inputs from the city of Puno and is the most eutrophic sector.
133 Measured transparency spans 1.5 m in near-shore turbid water to 16.5 m in
134 the open oligotrophic basin.

(a) Red de monitoreo y gradiente de transparencia

Estaciones IMARPE/ALT con match-up satélite-campo (2013–2024), coloreadas por transparencia media medida. El gradiente trófico es visible en el propio dato de campo.



(b) El esfuerzo de muestreo es muy desigual

Cada punto es una estación. 319 estaciones, pero la mayoría se visitó en muy pocas campañas, lo que limita el poder de cualquier análisis por estación

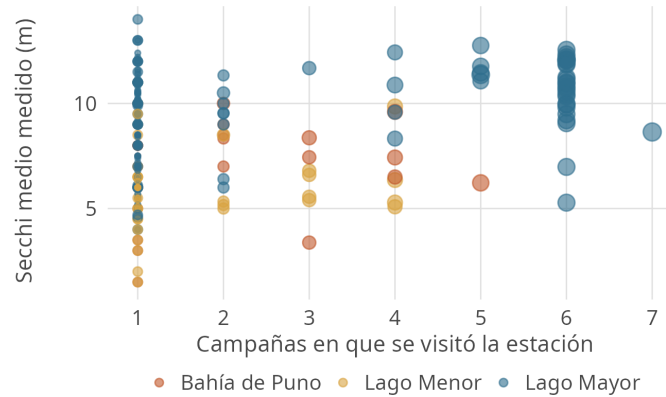


Figure 1: Study area and monitoring network. (a) IMARPE/ALT stations with a satellite–field match-up, coloured by mean measured Secchi depth over the Natural Earth lake boundary; the trophic gradient is evident in the field data alone. (b) Sampling effort per station: most stations were visited in few campaigns, which limits the power of any station-level analysis.

135 2.2. *In-situ data*

136 We used 881 field measurements of Secchi transparency, with co-measured
137 chlorophyll-*a*, suspended solids, temperature, pH, dissolved oxygen and nu-
138 trients, from the limnological monitoring campaigns of IMARPE (Instituto
139 del Mar del Perú) on Lake Titicaca, spanning 2011–2024 across 156 sta-
140 tions distributed over the whole lake. Of these, 734 include a valid Secchi
141 reading. The July 2019 campaign (71 stations) is published by Siguyro Ma-
142 mani and Franco (2022); the remaining years derive from the same IMARPE
143 programme. As a provenance check, our 2019 zone-level means reproduce
144 the published values almost exactly (Lago Mayor conductivity 1513.4 vs.
145 1513.5 $\mu\text{S cm}^{-1}$; Lago Menor chlorophyll-*a* 1.52 vs. 1.52 mg m^{-3}).

146 Three properties of this record constrain everything that follows, and we
147 state them here rather than in the limitations (Fig. 2).

148 First, the monitoring programme is itself almost entirely a dry-season
149 programme. The 2011–2024 record contains 13 campaigns with Secchi read-
150 ings, of which exactly one — December 2012, 72 measurements — falls in the
151 austral wet season (December–March). That campaign predates both sen-
152 sors used here: Landsat 8 became operational in May 2013 and Sentinel-2A
153 in June 2015. Consequently every match-up in our dataset falls between July
154 and October, and this is a property of the field programme rather than of
155 our processing: no choice of match-up window or cloud threshold can recover
156 wet-season pairs that have no contemporaneous imagery. We return to what
157 this means for operational use in Section 4.3.

158 Second, sampling is episodic: campaigns exist for 2013–2024 with no
159 campaigns in 2020, 2021, 2023.

160 Third, the reference measurement is coarse: 73 distinct Secchi values
161 occur in 812 match-ups, because disk readings are recorded to the nearest
162 half metre.

163 No synthetic or spectrally derived labels were used at any stage.

164 2.3. *Satellite data and cross-sensor harmonization*

165 Sentinel-2 MSI (harmonized surface reflectance) and Landsat 8/9 OLI
166 (Collection 2, Level-2; Vermote et al., 2016) were accessed through Google
167 Earth Engine (Gorelick et al., 2017). Clouds were masked with the Sentinel-
168 2 Scene Classification Layer plus cloud probability, and with the Landsat
169 QA_PIXEL bitmask (dilated cloud, cloud and shadow). The cirrus flag was
170 not applied, as it is near-permanently set at 3810 m and would mask the
171 entire archive. We retained the six bands common to both sensors (blue,
172 green, red, NIR, SWIR1, SWIR2).

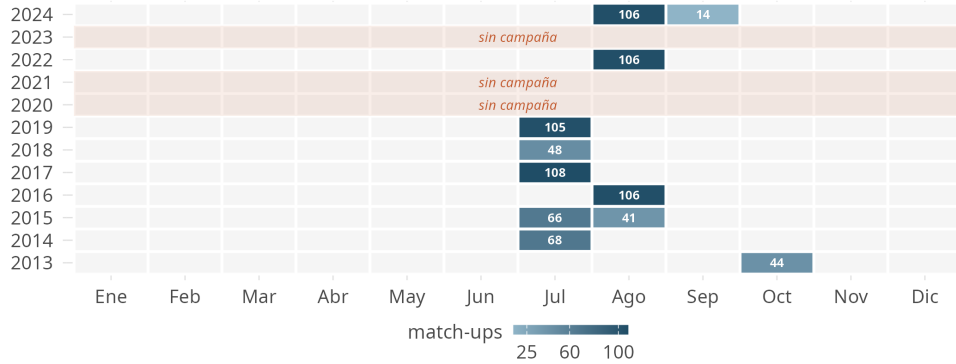
173 Landsat surface reflectance was first harmonized to Sentinel-2 equivalent
174 reflectance with the per-band linear coefficients of Roy et al. (2016), of the
175 form $\rho_{\text{MSI}} = a + b \rho_{\text{OLI}}$. Section 3.1 evaluates whether this transform is valid
176 over the water of Lake Titicaca and derives an alternative. For that evalu-
177 ation we identified every in-situ event observed by both sensors within the
178 match-up window — 270 paired station–date observations — and regressed
179 Sentinel-2 on harmonized Landsat reflectance band by band.

180 2.4. *Satellite–field match-ups*

181 For every in-situ measurement we extracted the cloud-masked median
182 reflectance within a 30 m buffer of the station coordinate, from imagery
183 acquired within ± 10 days. The 30 m buffer matches the coarser Landsat pixel

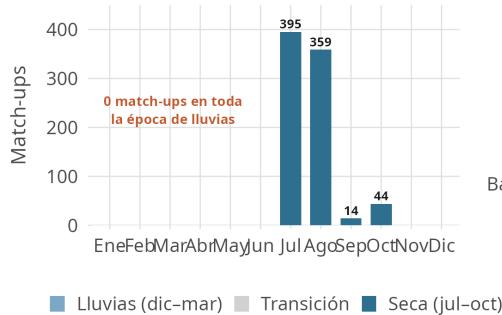
(a) Tres años sin campaña y ni un solo dato fuera de la época seca

Número de match-ups satélite-campo por año y mes. No existe dato de 2020, 2021 ni 2023, así que el hold-out «2022–2024» sólo contiene 2022 y 2024



(b) Sólo hay datos de época seca

Las floraciones de Bahía de Puno ocurren en época de lluvias, que el dataset no cubre en absoluto



(c) El gradiente trófico domina el Secchi

Sólo 73 valores distintos en 812 match-ups: las lecturas se anotan al medio metro, así que el RMSE de 1.86 m roza la granularidad de la propia medida de referencia

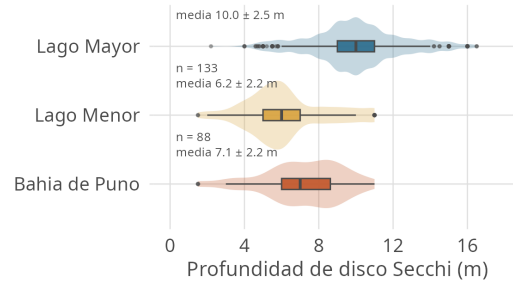


Figure 2: What the dataset actually contains. (a) Match-ups by year and month; three years have no campaign at all, so a hold-out described as “2022–2024” contains only 2022 and 2024. (b) Seasonal coverage: zero match-ups in the wet season, when blooms occur in Bahía de Puno. (c) Secchi distribution by trophic zone; the reference measurement is quantized at roughly half a metre.

184 and averages residual geolocation error and sensor noise over several pixels
185 while remaining in open water, away from shoreline mixing. The ± 10 -day
186 window balances the scarce cloud-free image availability at 3810 m against
187 the slow temporal change of transparency in this large, deep lake, and is
188 consistent with windows used in operational Z_{SD} retrieval (Kloiber et al.,
189 2002; Zhang et al., 2022).

190 This yielded 1002 match-ups (340 Sentinel-2, 662 Landsat), of which 812
191 have a Secchi value, across 143 stations and 120 campaign dates. Because
192 both sensors can observe the same field event, 272 match-ups are cross-sensor
193 pairs of a single in-situ measurement; collapsing them leaves 540 independent
194 events, and Section 3.2 verifies that accuracy is unchanged, confirming the
195 result is not an artefact of pseudo-replication.

196 2.5. Features and model

197 Twelve sensor-agnostic features were computed from the six common
198 bands: the bands themselves plus NDWI, NDTI and the ratios blue/green,
199 green/blue, red/green and blue/red. We trained a Random Forest regres-
200 sor (Breiman, 2001). Reflectance was scaled with a robust (median/IQR)
201 scaler fitted only on the training partition of each fold. All models were
202 implemented with scikit-learn (Pedregosa et al., 2011).

203 Model choice was settled empirically rather than assumed. We com-
204 pared Random Forest, Extremely Randomized Trees, XGBoost (Chen and
205 Guestrin, 2016) and LightGBM, each with hyperparameters optimized by
206 Optuna (Akiba et al., 2019) inside *nested* cross-validation — tuning on an
207 inner campaign-blocked split of each outer training partition, so no outer fold
208 ever informs its own hyperparameters — and benchmarked the resulting gain

209 against the spread across random seeds. Section 3.4 reports the outcome.

210 2.6. Validation design and null models

211 Rather than a single split, we evaluate a hierarchy of increasingly de-
212 manding designs (Roberts et al., 2017): random K -fold; GroupKFold blocked
213 by station; GroupKFold blocked by *campaign date*; GroupKFold blocked by
214 year; a temporal hold-out; and leave-one-zone-out, in which an entire trophic
215 zone is withheld.

216 Crucially, we do not assume which blocking unit matters. We benchmark
217 each design against null models that predict, respectively, the training-set
218 mean of the zone, of the station and of the campaign date. A blocking unit
219 is doing real work if, and only if, the corresponding null model collapses
220 under it. This diagnostic, rather than convention, determines our primary
221 design.

222 Performance is reported as out-of-fold R^2 , RMSE and MAE, and bench-
223 marked against a classical log-linear blue/green band-ratio algorithm in the
224 standard form $\ln Z_{SD} \sim \ln(\rho_{\text{blue}}/\rho_{\text{green}})$ (Kloiber et al., 2002), with predic-
225 tions bounded to the physically observable range, and against a multiband
226 linear model.

227 2.7. Uncertainty and interpretation

228 We attach a per-prediction interval using split-conformal prediction (An-
229 gelopoulos and Bates, 2023): within each fold, a calibration subset of the
230 training partition yields the $(1 - \alpha)$ quantile of absolute residuals, defining
231 a symmetric interval with finite-sample marginal coverage guarantees (Ro-
232 mano et al., 2019). Because a single nominal level can be met by chance, we

233 report empirical coverage at 50, 80, 90 and 95%, and also coverage condi-
234 tional on measured transparency, since marginal calibration does not imply
235 conditional calibration.

236 The model is interpreted with SHAP (Lundberg and Lee, 2017; Lundberg
237 et al., 2020), computed *out of fold* — each observation explained by the model
238 that did not see it — rather than on a model fitted to the whole dataset, which
239 inflates the apparent importance of memorized features. Trends in annual
240 median in-situ transparency were tested with Mann–Kendall and Sen’s slope
241 (Mann, 1945; Sen, 1968) on the full 2011–2024 in-situ record, not on the
242 match-up subset.

243 3. Results

244 3.1. Land-derived harmonization is invalid over oligotrophic water

245 The Roy et al. (2016) transforms carry additive intercepts of 0.0116–
246 0.0448 in reflectance units. Over Lake Titicaca the native Landsat over-
247 water signal is far smaller than this (Table 1, Fig. 3a). In the near-infrared
248 the median native reflectance is effectively zero, so the intercept exceeds the
249 signal by a factor of 69.8; in SWIR1 the factor is , in red 10.6, and even
250 in blue — the brightest water band — 2.5. The harmonized Landsat bands
251 are consequently dominated by a constant inherited from a land calibration:
252 their medians exceed the corresponding Sentinel-2 medians by 1.6-fold in
253 green and 15.4-fold in the near-infrared.

254 A harmonization that succeeded would make the two sensors statistically
255 indistinguishable over the same water. They are not. A Random Forest clas-
256 sifier trained to identify the source sensor from the harmonized features alone

Table 1: Effect of the Roy et al. (2016) OLI→MSI transform over Lake Titicaca water. Native OLI is the Collection-2 surface reflectance recovered by inverting the transform. Reflectance is dimensionless.

Band	Roy intercept a	Native OLI median	Intercept / signal	Harmonized LS median	Harmonized / S2 median
Blue	0.0183	+0.0072	2.5	0.0247	1.74
Green	0.0123	+0.0089	1.4	0.0206	1.61
Red	0.0123	+0.0013	10.6	0.0134	2.66
NIR	0.0448	−0.0006	69.8	0.0443	15.42
SWIR1	0.0306	+0.0013	23.2	0.0317	13.34
SWIR2	0.0116	+0.0013	8.2	0.0129	5.70

reaches 99.8% accuracy against a 66.3% majority-class baseline (Fig. 3b); restricting it to the visible bands gives 97.9%, and to scale-invariant band ratios 99.5%. This is a classifier two-sample test (Friedman, 2003; Lopez-Paz and Oquab, 2017): if two samples come from the same distribution, no classifier can separate them above chance. Sensor identity therefore remains fully encoded in the harmonized reflectance, and any model trained on the pooled archive can read it. The test is not new, but to our knowledge it has not been used as a harmonization acceptance check; it costs one model fit and it is decisive.

Refitting locally on the 270 station–date events observed by both sensors gives water-specific coefficients (Table 2). These restore the spectral signature almost exactly — the median Landsat/Sentinel-2 ratio falls from 15.4 to about 1.0 across all six bands (Fig. 3d) — but per-observation agreement

remains uneven (Fig. 3c). Green recovers usable agreement ($r = 0.71$), and NIR and SWIR reach $r \approx 0.6$. Blue does not: $r = 0.26$. This is the most consequential result of the section, because SHAP attributes 72.6% of the retrieval to green/blue bands and ratios (Section 3.7). The band on which the retrieval most depends is the band on which the two sensors least agree.

Table 2: Water-specific LS→S2 recalibration fitted on 270 paired station–date observations, of the form $\rho_{S2} = a + b \rho_{LS, Roy}$.

Band	a	b	Pearson r	Agreement
Blue	+0.0084	0.258	0.26	unusable
Green	−0.0009	0.665	0.71	good
Red	−0.0005	0.427	0.52	weak
NIR	−0.0250	0.631	0.62	acceptable
SWIR1	−0.0223	0.770	0.64	acceptable
SWIR2	−0.0057	0.615	0.60	acceptable

3.2. The campaign, not the station, is the blocking unit

Table 3 reports the retrieval under each validation design alongside the null models (Fig. 4). Blocking by station changes nothing relative to a random split (both $R^2 = 0.596$), which under the usual reading would be taken as evidence of no spatial leakage. The null models show why that reading is wrong: under campaign-blocked folds the station-mean null still attains $R^2 = 0.208$ and the zone-mean null $R^2 = 0.287$, whereas the campaign-date-mean null collapses to -0.012 . Dependence in this dataset lives between campaigns, not between stations, because a single campaign samples the

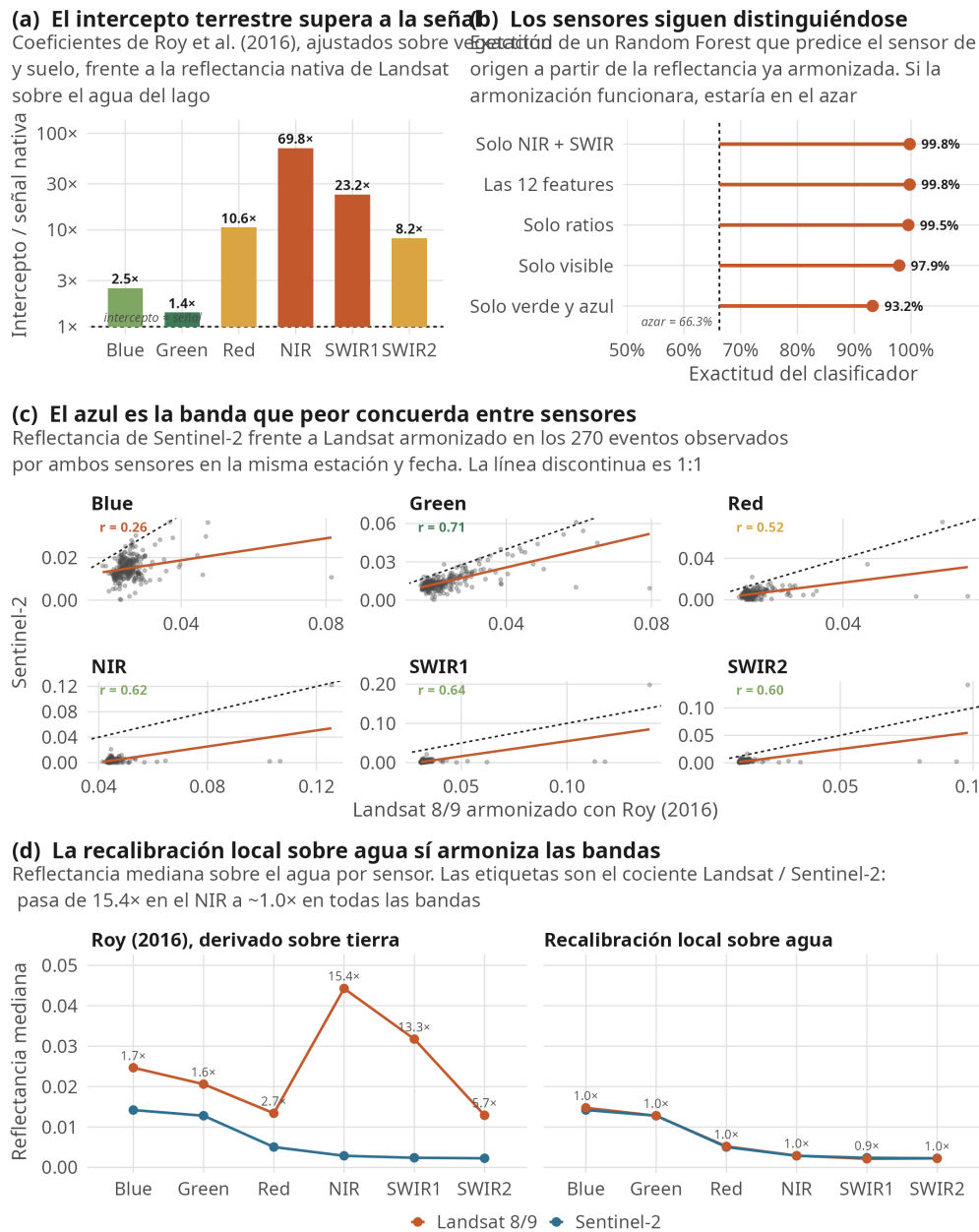


Figure 3: Why land-derived harmonization fails over clear water. (a) The Roy intercept relative to the native over-water Landsat signal, by band. (b) Accuracy of a classifier predicting the source sensor from harmonized features; under a successful harmonization it would sit at chance. (c) Sentinel-2 versus harmonized Landsat reflectance on the 270 paired station-date events; blue agrees worst. (d) Median spectral signature before and after local water recalibration; labels are the Landsat/Sentinel-2 ratio.

284 whole lake within a few days under one set of atmospheric and limnological
 285 conditions. Station-blocked folds leave that structure intact.

286 Under campaign-blocked validation — our primary design — the Random
 287 Forest attains $R^2 = 0.574$ and RMSE 1.86 m over 812 match-ups, essentially
 288 unbiased (-0.02 m). Blocking by year is stricter still ($R^2 = 0.495$). Collaps-
 289 ing cross-sensor pseudo-replicates to 540 independent events leaves accuracy
 290 unchanged ($R^2 = 0.571$).

Table 3: Validation design, null models and blocking units (Random Forest, $n = 812$ unless noted). A blocking unit is effective only if the corresponding null model collapses under it.

	Setting	R^2	RMSE (m)
Null models	Zone-mean	0.287	2.40
	Station-mean	0.208	2.53
	Campaign-date-mean	-0.012	2.86
Blocking unit	Random K -fold	0.596	1.81
	By station	0.596	1.81
	By campaign date	0.574	1.86
	By year	0.495	2.02
	One record per event ($n = 540$)	0.571	1.77
Zone extrapolation	Leave out Lago Menor ($n = 133$)	0.247	1.90
	Leave out Bahía de Puno ($n = 88$)	0.036	2.11
	Leave out Lago Mayor ($n = 591$)	-0.047	2.52
Temporal	Train ≤ 2019 , test 2022 & 2024 ($n = 226$)	0.507	2.63

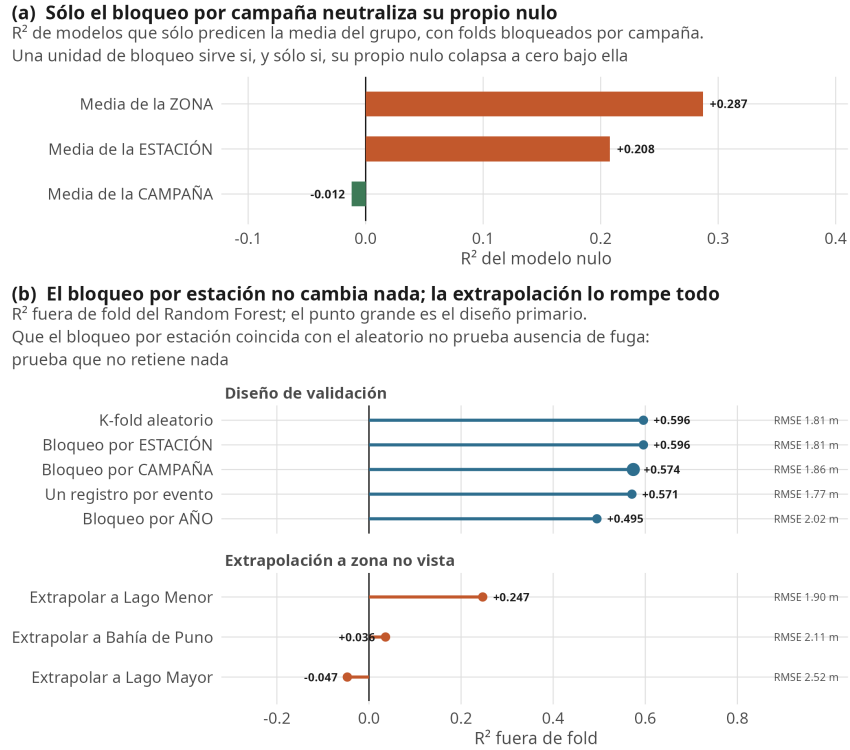


Figure 4: Diagnosing the blocking unit. (a) Null models evaluated under campaign-blocked folds: only the campaign-mean null collapses, so only campaign blocking is doing work. (b) Out-of-fold R^2 under each validation design and under extrapolation to an unseen trophic zone. That station blocking matches the random split is not evidence of no leakage; it is evidence that station blocking retains nothing.

291 *3.3. Retrieval accuracy and benchmarks*

292 Against this design the classical blue/green band ratio attains $R^2 = 0.097$
 293 (RMSE 2.71 m) and a multiband linear model $R^2 = 0.323$ (RMSE 2.34 m)
 294 (Table 4, Fig. 5). We report one methodological caution alongside these fig-
 295 ures. An unbounded implementation of the same band-ratio algorithm —
 296 log-transformed and back-transformed without constraining predictions to
 297 the observable range — produces a single prediction of 143.5 m against an
 298 observed maximum of 16.5 m, which alone drives R^2 to -2.458 and RMSE to
 299 5.29 m (Fig. 5d). Benchmarks of classical algorithms reported without phys-
 300 ical bounds should be treated with caution: a single unconstrained extrapo-
 301 lation manufactures an arbitrarily large advantage for the machine-learning
 302 model.

Table 4: Head-to-head benchmark, campaign-blocked GroupKFold, $n = 812$.

Algorithm	R^2	RMSE	MAE	Bias
Classical blue/green, unbounded ^a	-2.458	5.29	1.91	-0.19
Classical blue/green, bounded ^b	0.097	2.71	2.13	-0.46
Multiband linear	0.323	2.34	1.78	-0.01
Random Forest (this study)	0.574	1.86	1.44	-0.02

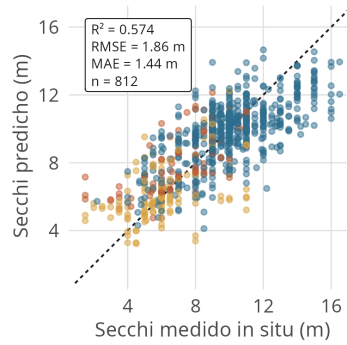
RMSE, MAE and bias in metres.

^a back-transformed without bounding predictions; see text.

^b standard form of Kloiber et al. (2002).

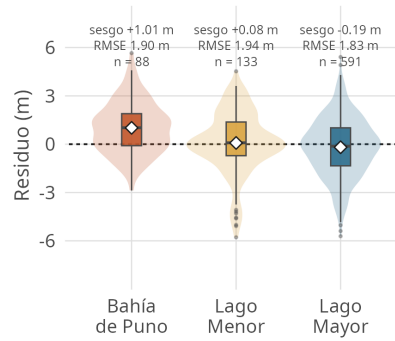
(a) Predicho frente a medido

Random Forest, validación bloqueada por campaña. La discontinua es 1:1



(b) Sobreestima donde más importa

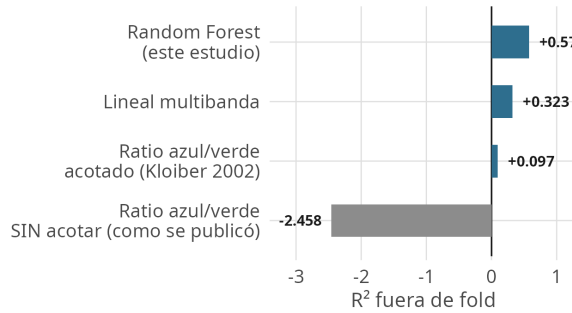
Residuos (predicho – medido) por zona. El rombo es la media. Bahía de Puno, la zona más eutrófica, +1.0 m



• Bahía de Puno • Lago Menor • Lago Mayor

(c) El baseline estaba mal implementado

Mismo diseño de validación para todos. El baseline «publicado» deshacía el log sin acotar; una única predicción de 143 m hundía su R^2



(d) Una predicción imposible basta

Ratio clásico en escala logarítmica. El punto de 143 m es el que producía el $R^2 = -2.46$

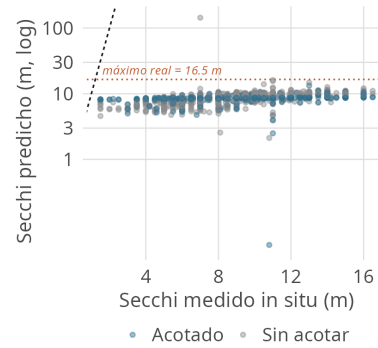


Figure 5: Retrieval under the primary design. (a) Out-of-fold predicted versus measured Secchi depth. (b) Residuals by trophic zone; the retrieval overestimates in eutrophic Bahía de Puno. (c) Benchmark against classical and linear baselines. (d) The unbounded classical implementation in log scale: one impossible prediction destroys the metric.

3.4. *Model family and tuning do not change the picture*

We compared four learner families, each tuned by Optuna inside nested cross-validation, against target transformations and reduced feature sets (Table 5). The best configuration — an equal-weight ensemble of the four tuned models — reached $R^2 = 0.583$, an improvement of 0.009 over the untuned Random Forest. That gain does exceed the spread across random seeds (0.003 in R^2), so it is not noise; but it corresponds to 0.02 m of RMSE on a lake whose reference measurement is recorded to the nearest half metre, and it changes no conclusion in this article. Neither log-transforming the target nor tuning individual learners helped more than that, and restricting the feature set cost accuracy.

We therefore retain the untuned Random Forest as the reported model: it is simpler, faster and easier to reproduce, and reporting a tuned ensemble for a 0.009 gain would overstate the precision of the product. The ceiling here is set by the data — a dry-season-only record, a reference measurement quantized at half a metre, and a target whose variance is dominated by a three-level trophic gradient — not by the choice of learner.

3.5. *Where the retrieval stops working*

Withholding an entire trophic zone collapses the retrieval (Table 3): R^2 falls to 0.247 for Lago Menor, 0.036 for Bahía de Puno and -0.047 for Lago Mayor. The model interpolates within optical regimes it has seen and does not extrapolate to a water type absent from training. On the temporal hold-out (train ≤ 2019 , test 2022 and 2024) it retains $R^2 = 0.507$, though 57 of the 69 test stations also appear in training (82.6%), so this measures temporal rather than fully independent generalization.

Table 5: Model selection under nested cross-validation (Optuna inside an inner campaign-blocked split of each outer training partition). Selected configurations; the full grid is in the repository.

Model	Features	R^2	RMSE (m)
Ensemble of the four tuned models	all 12	0.583	1.84
LightGBM, tuned	all 12	0.581	1.84
XGBoost, tuned	all 12	0.577	1.85
Random Forest, tuned	all 12	0.576	1.85
Random Forest, untuned (reported)	all 12	0.574	1.86
Extremely Randomized Trees, untuned	all 12	0.573	1.86
XGBoost, tuned, $\log Z_{SD}$	all 12	0.561	1.89
LightGBM, tuned	visible only	0.544	1.92
LightGBM, tuned	ratios only	0.488	2.04

Seed-to-seed spread of the untuned Random Forest: R^2 range 0.003 over five seeds.

328 Error is unevenly distributed (Table 6). The retrieval carries a positive
 329 bias of 1.01 m in Bahía de Puno, the most eutrophic and most vulnerable
 330 zone, because the clear-water majority pulls predictions upward.

Table 6: Out-of-fold error by trophic zone, campaign-blocked validation.

Zone	R^2	RMSE (m)	Bias (m)	n
Bahía de Puno	0.221	1.90	+1.01	88
Lago Menor	0.211	1.94	+0.08	133
Lago Mayor	0.448	1.83	−0.19	591

331 Two features of the harmonized archive should be read together with these
 332 numbers. Removing the two bands most corrupted by the land-derived in-
 333 tercepts lowers R^2 only from 0.574 to 0.541; using scale-invariant ratios alone
 334 gives 0.468; and applying our water-specific recalibration gives 0.518. Train-
 335 ing on each sensor separately gives 0.563 (Sentinel-2, $n = 274$) and 0.549
 336 (Landsat, $n = 538$). The pooled model’s small advantage over these vari-
 337 ants is consistent with it exploiting sensor identity as an additional covariate
 338 rather than with genuine cross-sensor synergy.

339 3.6. Calibrated per-prediction uncertainty

340 Split-conformal intervals are well calibrated *marginally* at every nominal
 341 level tested: empirical coverage is 50.9% at 50% nominal, 79.1% at 80%,
 342 89.4% at 90% and 94.3% at 95%, a maximum calibration error of 0.94 per-
 343 centage points (Fig. 6a).

344 They are not calibrated *conditionally*, and this matters more for opera-
 345 tional use than the marginal result. Coverage falls to 71.3% in the clearest

346 water ($Z_{SD} > 12.5$ m) and 79.5% in the most turbid ($Z_{SD} < 5$ m), while ex-
 347 ceeding the nominal level in the middle of the range (Fig. 6c). The mechanism
 348 is structural rather than incidental: split conformal derives a single global
 349 quantile of absolute residuals, so the interval has constant width (6.12 m at
 350 90%) everywhere and cannot adapt to the local difficulty of the retrieval. It
 351 therefore buys its marginal guarantee by over-covering in the well-sampled
 352 middle of the range and under-covering at both extremes — which are pre-
 353 cisely where management decisions are made. A locally adaptive variant such
 354 as conformalized quantile regression (Romano et al., 2019) would be required
 355 for reliable conditional coverage, and we recommend reporting coverage con-
 356 ditional on the predicted value as standard practice whenever conformal in-
 357 tervals accompany an operational product.

358 For a lake whose transparency spans 1.5–16.5 m, a ± 3 m interval is in-
 359 formative for zone-scale contrasts and multi-metre change, and is not infor-
 360 mative for sub-metre differences.

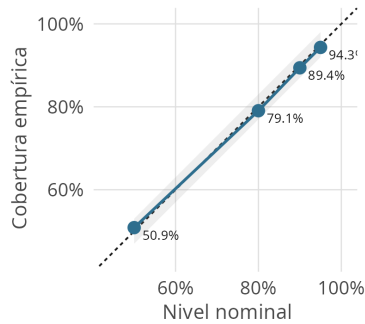
361 3.7. *Interpretation and retrievability*

362 Out-of-fold SHAP attributes 72.6% of the retrieval to green and blue
 363 bands and their ratios, 14.4% to NIR and SWIR, and the remainder to
 364 other indices (Fig. 7a). Green/blue alone accounts for 37%. The dependence
 365 is monotone and of the sign expected from water-clarity optics (Fig. 7b):
 366 a higher green-to-blue ratio pushes the prediction toward less transparent
 367 water. The retrieval is therefore grounded in optics rather than location
 368 memorization — while resting on the band pair with the worst cross-sensor
 369 agreement.

370 With real labels and the same campaign-blocked design, no other vari-

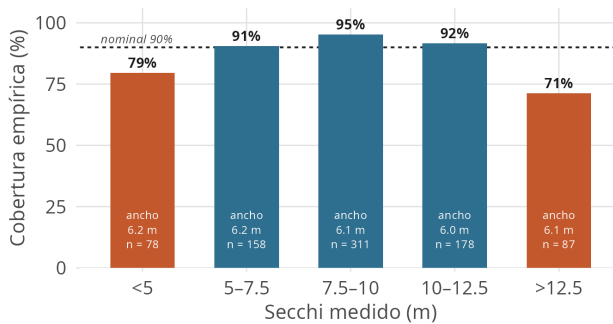
(a) Calibrados... en promedio

Cobertura empírica fuera de fold frente a la nominal. La banda gris es ± 3 puntos que la anchura es constante (~ 6.1 m) y el intervalo no puede adaptarse: infracubre en el agua más clara



(c) Pero no en los extremos del rango

El conformal split usa un único cuantil global, así a la nominal. La banda gris es ± 3 puntos que la anchura es constante (~ 6.1 m) y el intervalo no puede adaptarse: infracubre en el agua más clara



(b) El intervalo al 90% frente a la medida real

Match-ups ordenados por transparencia medida. La línea es la predicción, la banda el intervalo conformal al 90%. 86 de 812 puntos quedan fuera

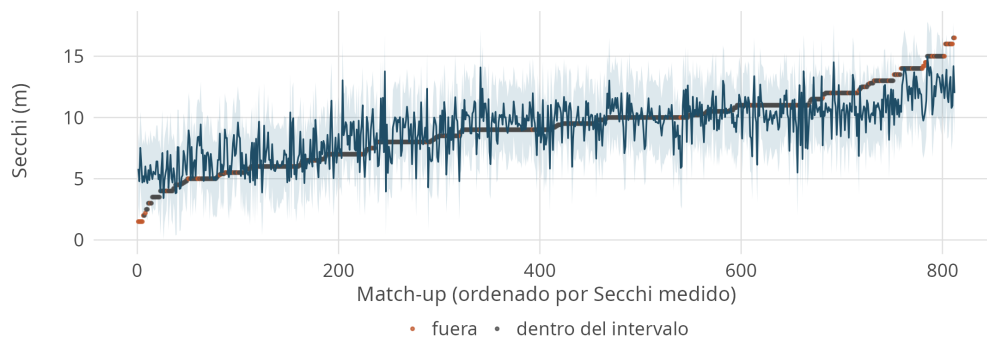


Figure 6: Conformal prediction intervals under campaign-blocked validation. (a) Empirical versus nominal coverage: marginally calibrated across the 50–95% range. (b) The 90% interval against the measured value. (c) Coverage conditional on measured transparency. Because split conformal uses a single global residual quantile, the interval width is constant (≈ 6.12 m) and cannot adapt, so the marginal guarantee is bought by over-covering in the middle of the range and under-covering at both extremes.

371 able is retrievable (Fig. 7c): chlorophyll-*a* reaches $R^2 = 0.012$, suspended
 372 solids 0.056, in-situ water temperature 0.113, and Landsat thermal surface
 373 temperature 0.020 ($r = 0.14$, $n = 662$). Their in-situ ranges — chlorophyll-*a*
 374 0–4.6 mg m⁻³, suspended solids mostly below 10 mg L⁻¹ — lie below the op-
 375 tical and thermal detection limits of these sensors. We subject the negative
 376 claim to the same design as the positive one, and report it to caution against
 377 applying chlorophyll algorithms (Gitelson et al., 2009; Mishra and Mishra,
 378 2012) or thermal retrievals (Ruiz-Verdú et al., 2016) in clear high-altitude
 379 water without local validation.

380 *3.8. Temporal patterns and the fragility of trend claims*

381 Mann–Kendall tests on annual median in-situ transparency over the full
 382 2011–2024 record show no statistically significant trend in any zone (Bahía
 383 de Puno: Sen slope 0.16 m yr⁻¹, $p = 0.241$; Lago Mayor: 0.25 m yr⁻¹,
 384 $p = 0.138$; Fig. 8a).

385 That conclusion is fragile in a way worth reporting explicitly. Restricting
 386 the series to 2013 onwards — the natural choice if trends were computed
 387 on the satellite match-up subset, whose record begins with Landsat 8 —
 388 turns both Bahía de Puno ($p = 0.017$) and Lago Mayor (Sen +0.42 m yr⁻¹,
 389 $p = 0.012$) significantly positive. Across start years from 2011 to 2014, 5
 390 of 12 zone–start-year combinations change verdict (Fig. 8b). The year 2013
 391 has the lowest annual medians on record, so beginning there manufactures
 392 an upward trend. With 9–11 annual values and three gap years the test has
 393 low power, and the correct reading is absence of evidence, not evidence of
 394 stability. We recommend that trend analyses in campaign-based lake records
 395 report this sensitivity as a matter of course.

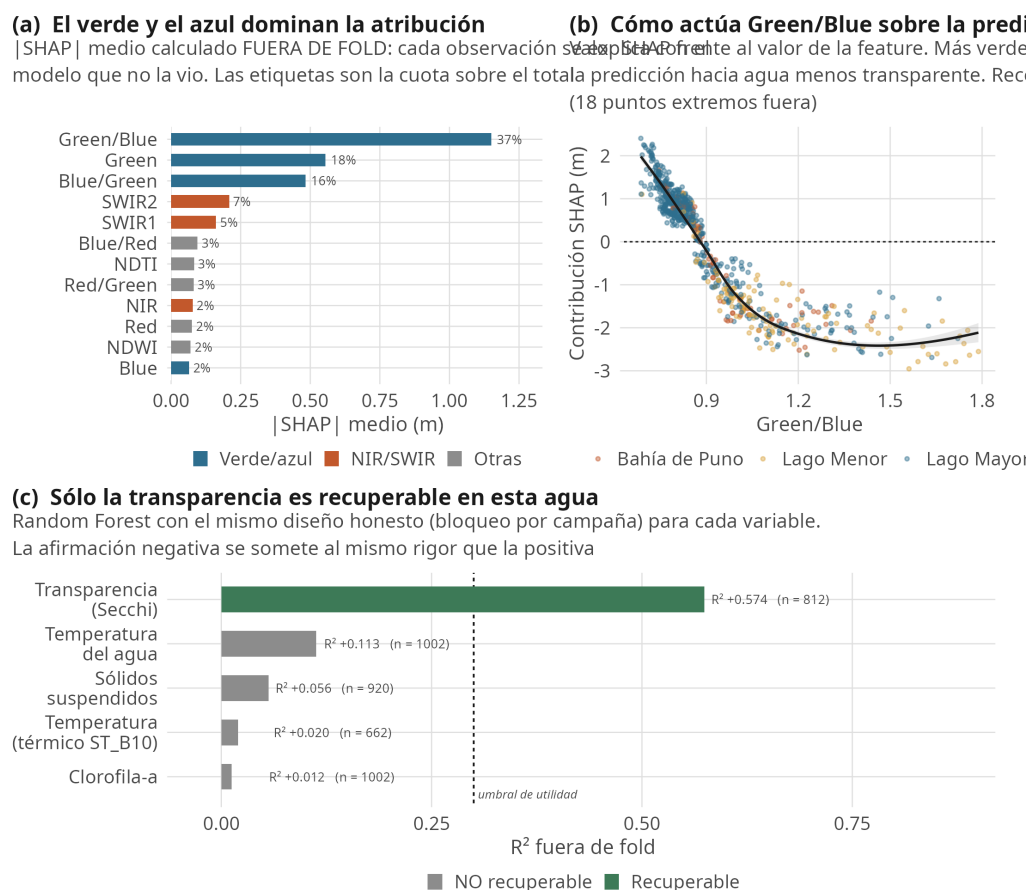
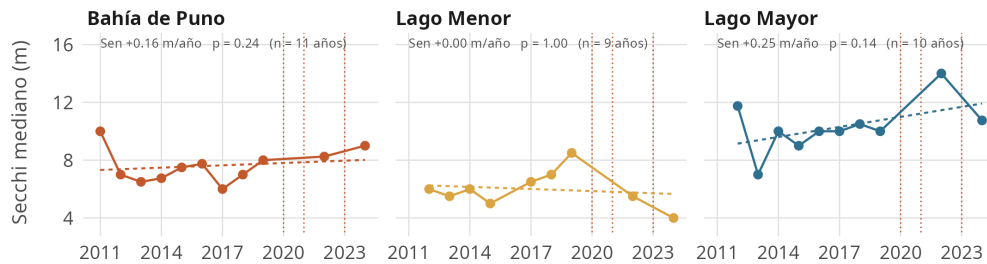


Figure 7: Interpretation and operational envelope. (a) Out-of-fold SHAP importance; green and blue dominate. (b) SHAP dependence on the leading feature, by trophic zone. (c) Retrievalability of each water-quality variable under the same campaign-blocked design; only transparency clears a useful threshold.

(a) Ninguna zona muestra tendencia significativa en el registro completo

Mediana anual de transparencia in situ (2011–2024, 734 lecturas). Las líneas punteadas verticales marcan los años sin campaña: 2020, 2021 y 2023



(b) El veredicto lo fija el año en que empieza la serie, no el lago

Mismo test de Mann-Kendall, mismos datos, distinto año de inicio. Arrancar en 2013 —el año con las medianas más bajas del registro— convierte dos zonas en «significativamente más transparentes»

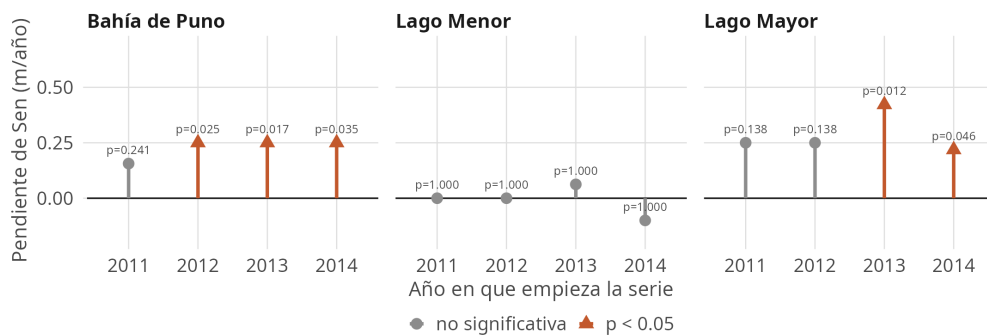


Figure 8: Transparency trends and their fragility. (a) Annual median in-situ transparency by zone over the full 2011–2024 record, with Mann–Kendall/Sen statistics; no trend is significant. (b) The same test with different start years: beginning the series in 2013, the lowest-median year on record, turns two zones significantly positive.

396 4. Discussion

397 4.1. Implications for harmonized multi-mission archives

398 The virtual-constellation approach (Claverie et al., 2018; Pahlevan et al.,
399 2019) has become the default for multi-decadal lake monitoring, and the Roy
400 et al. (2016) coefficients are its most reused transform. Our results place
401 a boundary on where that default is safe. Those coefficients were fitted on
402 terrestrial targets and carry additive intercepts appropriate to surfaces one
403 to two orders of magnitude brighter than clear water. Over Lake Titicaca the
404 intercept exceeds the native over-water signal by a factor of 69.8 in the near-
405 infrared and by 2.5 even in blue, and the two sensors remain 99.8% separable
406 after the transform. That water reflectance from these products is unreliable
407 is already documented (Pahlevan et al., 2019; Masek et al., 2021); what our
408 paired observations add is the magnitude — the problem is not confined
409 to the near-infrared, and it is large enough to make the harmonized bands
410 function as sensor labels for a machine-learning model — and the finding that
411 local recalibration repairs the aggregate spectral signature without repairing
412 per-observation agreement in blue.

413 The practical consequence is direct. A machine-learning model trained
414 on a pooled, land-harmonized archive can achieve apparently good accuracy
415 while having learned two sensor-specific mappings joined by a readable sensor
416 label. That is not a harmonized retrieval, and it will not behave as one when
417 applied to a scene from either sensor alone. The diagnostic is cheap and we
418 recommend it as routine: train a classifier to predict the source sensor from
419 the harmonized features and report its accuracy against the majority-class
420 baseline. If the sensors remain separable, the harmonization has not done

421 its job. For clear oligotrophic lakes we further recommend deriving water-
422 specific coefficients from same-day paired observations, reporting per-band
423 agreement, and — where blue-band agreement is as poor as we find here —
424 reporting single-sensor retrievals alongside any pooled product.

425 *4.2. Implications for validation design*

426 Blocking by station is the standard remedy for spatial autocorrelation
427 in this literature, and in our data it changes nothing: random and station-
428 blocked folds give identical accuracy. Read conventionally, that would be
429 reassuring. The null models show it is not. Under campaign-blocked folds
430 the station-mean null still explains 21% of variance and the zone-mean null
431 29%, while the campaign-date-mean null explains none. In a campaign-based
432 monitoring programme the whole lake is sampled within a few days under
433 shared atmospheric and limnological conditions, so observations cluster by
434 campaign far more strongly than by location; station-blocked folds leave that
435 structure entirely intact.

436 The general point is that the blocking unit should be diagnosed, not
437 assumed. Reporting null models under each candidate design costs little
438 and identifies the unit that carries the dependence. This applies well beyond
439 Lake Titicaca: most large-lake monitoring programmes sample in campaigns,
440 so the same mismatch between the conventional blocking unit and the real
441 dependence structure is likely to be widespread.

442 For our retrieval, the honest figure is $R^2 = 0.574$ (RMSE 1.86 m) under
443 campaign-blocked validation, against 0.287 for a zone-mean null: the model
444 roughly doubles the skill available from knowing only which basin a sample
445 came from. That is a real but modest contribution, and stating it that way

446 is more useful to a manager than an unqualified R^2 .

447 4.3. Operational envelope for Lake Titicaca

448 Our RMSE of 1.86 m spans roughly 12% of the observed Z_{SD} range
449 and sits below the published range for Z_{SD} retrieval ($R^2 \approx 0.6$ – 0.9 ; Zhang
450 et al., 2022; Lee et al., 2015), obtained here under a predominantly clear
451 oligotrophic regime and a stricter validation design. It is also approaching
452 the granularity of the reference measurement itself: disk readings are recorded
453 to the nearest half metre, giving only 73 distinct values across 812 match-ups.

454 Four constraints bound operational use. The retrieval carries a 1.01 m
455 positive bias in Bahía de Puno — the most eutrophic and most vulnerable
456 zone — so mapped values there must be read with their intervals and con-
457 firmed by field sampling. Leave-one-zone-out results show the model cannot
458 extrapolate to an unsampled optical regime; integration must be incremen-
459 tal, each new campaign both validating the current map and extending the
460 training set. The retrieval is calibrated and validated for the austral dry sea-
461 son only, and the reason is worth stating precisely because it determines the
462 remedy. It is not that wet-season imagery is unusable at 3810 m, nor that our
463 match-up criteria excluded it: it is that the field programme has produced
464 a single wet-season campaign in fourteen years, and that campaign predates
465 both satellites. Wet-season blooms in Bahía de Puno — the phenomenon a
466 transparency product would most usefully warn about — are therefore en-
467 tirely unrepresented, and the model should not be applied between December
468 and March.

469 This converts the limitation into a specific, actionable recommendation
470 for the monitoring authorities. Extending the protocol to the wet season does

not require a better algorithm; it requires field data that do not yet exist. A single additional wet-season campaign per year, timed to any cloud-free window and covering the three trophic zones, would within a few years supply enough independent campaign-blocks to calibrate and validate a wet-season retrieval under the same design used here. Because the binding constraint is sampling rather than sensing, this is among the cheapest available improvements to Lake Titicaca’s water-quality monitoring capability. And because conformal coverage degrades at both extremes of the transparency range, the nominal 90% interval should be treated as optimistic in the clearest and most turbid water.

Within those bounds the product is useful to the regional monitoring authorities (IMARPE, ALT, OEFA, and the binational management framework). A calibrated ± 3 m interval resolves zone-scale contrasts and multi-metre change, which is the scale at which clarity loss in Bahía de Puno would be actionable, and a single transboundary product under one calibration supports comparable assessment across the Peru–Bolivia boundary where coordinated field monitoring is logistically difficult. It does not resolve sub-metre change and should not be presented as doing so.

4.4. *Limitations*

Beyond the constraints already stated, three limitations bound this work. The in-situ record is episodic, with no campaigns in 2020, 2021, 2023 and only one wet-season campaign in fourteen years, so the temporal hold-out rests on two test years, the trend analysis on 9–11 annual values, and the seasonal envelope on none. Transparency above approximately 15 m saturates the optical signal. And this is a single lake: the two diagnostics we propose are

496 general, but the water-specific coefficients in Table 2 are local to Titicaca
497 and must be refitted elsewhere.

498 5. Conclusions

499 Using 1002 satellite–field match-ups over twelve years, we show that two
500 practices imported from the turbid-water remote-sensing literature fail in a
501 clear high-altitude large lake. Cross-sensor harmonization coefficients derived
502 over land leave Sentinel-2 and Landsat 99.8% separable over oligotrophic
503 water, because their additive intercepts exceed the native over-water signal
504 by up to a factor of 69.8; and validation blocked by monitoring station leaves
505 the dominant dependence — between campaigns — entirely intact.

506 Correcting both, a Random Forest retrieves Secchi depth with $R^2 = 0.574$
507 and RMSE 1.86 m, roughly double the skill of a zone-mean null, with confor-
508 mal prediction intervals that are calibrated marginally but under-cover at the
509 extremes of the transparency range. Skill collapses under extrapolation to an
510 unsampled trophic zone; chlorophyll-*a*, suspended solids and thermal tem-
511 perature are not retrievable at all; and no transparency trend is significant
512 once the full in-situ record is used rather than the satellite-era subset.

513 We therefore recommend two routine checks for multi-mission lake water-
514 quality products, neither of which requires new methodology: report the
515 accuracy of a classifier two-sample test on the harmonized features, and re-
516 port null models under each candidate validation blocking unit. Both cost
517 one model fit, and both distinguish genuine optical retrieval from sensor la-
518 belling and sampling design. For Lake Titicaca the resulting product is a
519 dry-season, zone-scale transparency indicator with honest ± 3 m intervals —

520 less than has sometimes been claimed for such lakes, and more than the
521 region has had.

522 **CRedit authorship contribution statement**

523 **Dina Maribel Yana-Yucra:** Conceptualization, Methodology, Inves-
524 tigation, Data curation, Writing – original draft. **Richar Andre Vilca-**
525 **Solorzano:** Methodology, Software, Formal analysis, Validation, Visualiza-
526 tion, Writing – original draft, Writing – review & editing. **Fred Torres-**
527 **Cruz:** Supervision, Resources, Writing – review & editing. **Vladimiro**
528 **Ibañez:** Supervision, Validation, Writing – review & editing.

529 **Declaration of competing interest**

530 The authors declare that they have no known competing financial inter-
531 ests or personal relationships that could have appeared to influence the work
532 reported in this paper.

533 **Data availability**

The satellite–field match-up datasets and the complete analysis pipeline, including the scripts that generate every number, table and figure in this article, are openly available at:

534 <https://github.com/Andre031222/harmonized-titicaca-transparency>



Repository

Raw imagery is freely available via Google Earth Engine; the in-situ measurements derive from published IMARPE/ALT and OEFA monitoring records.

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